

ADDRESSING A BLIND SPOT IN A TOPOLOGICAL DATA ANALYSIS–BASED DATA POISONING DETECTOR

Christian Dane Beels, Joseph Dorta, Ph.D.

Department of Mathematical Sciences, United States Military Academy, West Point, NY

Motivation

Data poisoning can insert attack-derived traffic with benign labels, shifting a learned decision boundary before deployment. Monkam et al. [1] propose an upstream TDA–OPTICS filter to identify poisoned packets before model fitting.

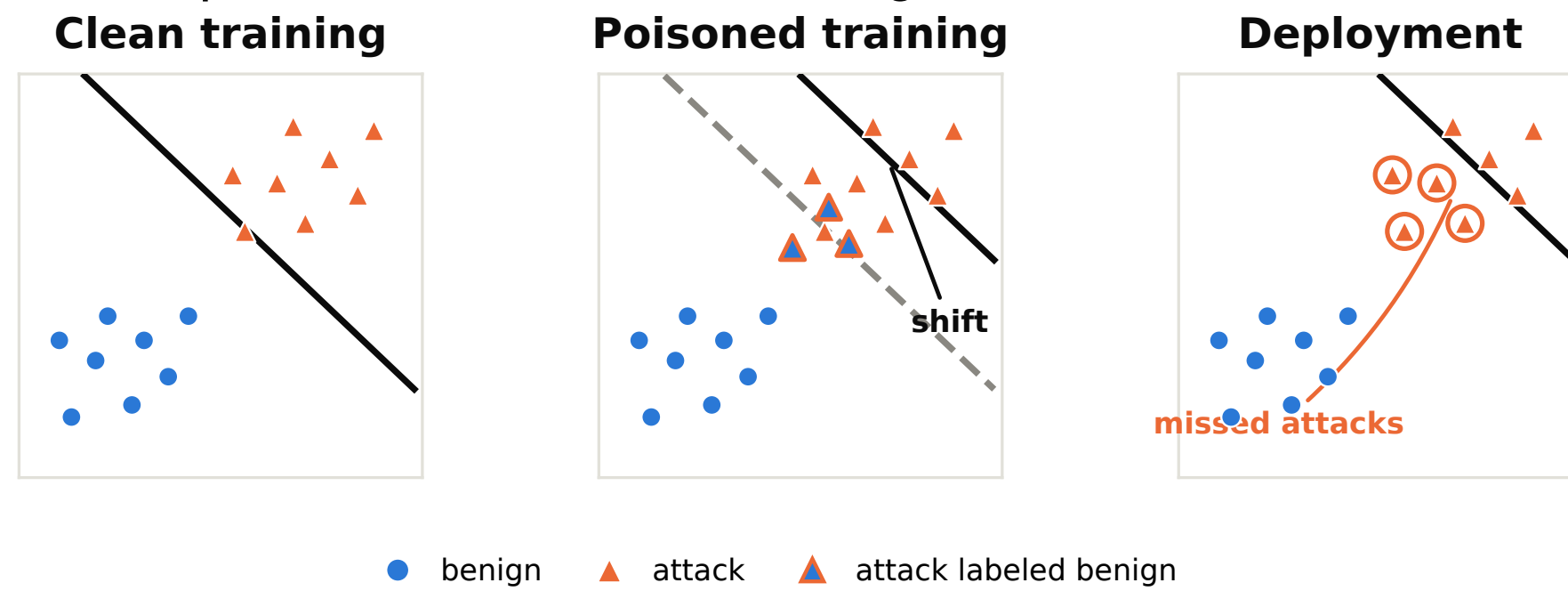


Figure 1. Data poisoning can cause mislabeling, which can shift a learned boundary and degrade model performance.

Persistent Homology

Persistent homology tracks connected components (H_0) and loops (H_1) across filtration scales; long intervals encode durable structure.

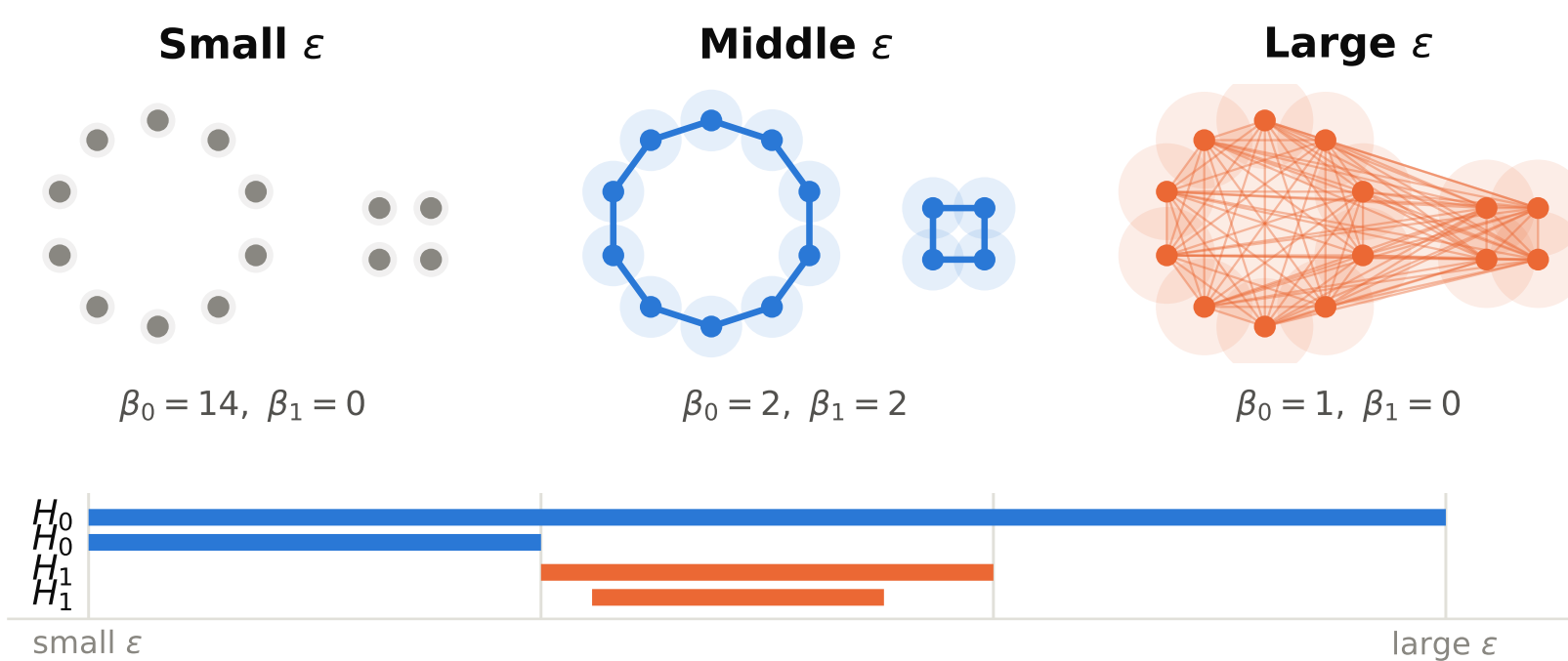


Figure 2. Point-cloud neighborhoods and the corresponding H_0/H_1 barcode as the scale ϵ increases.

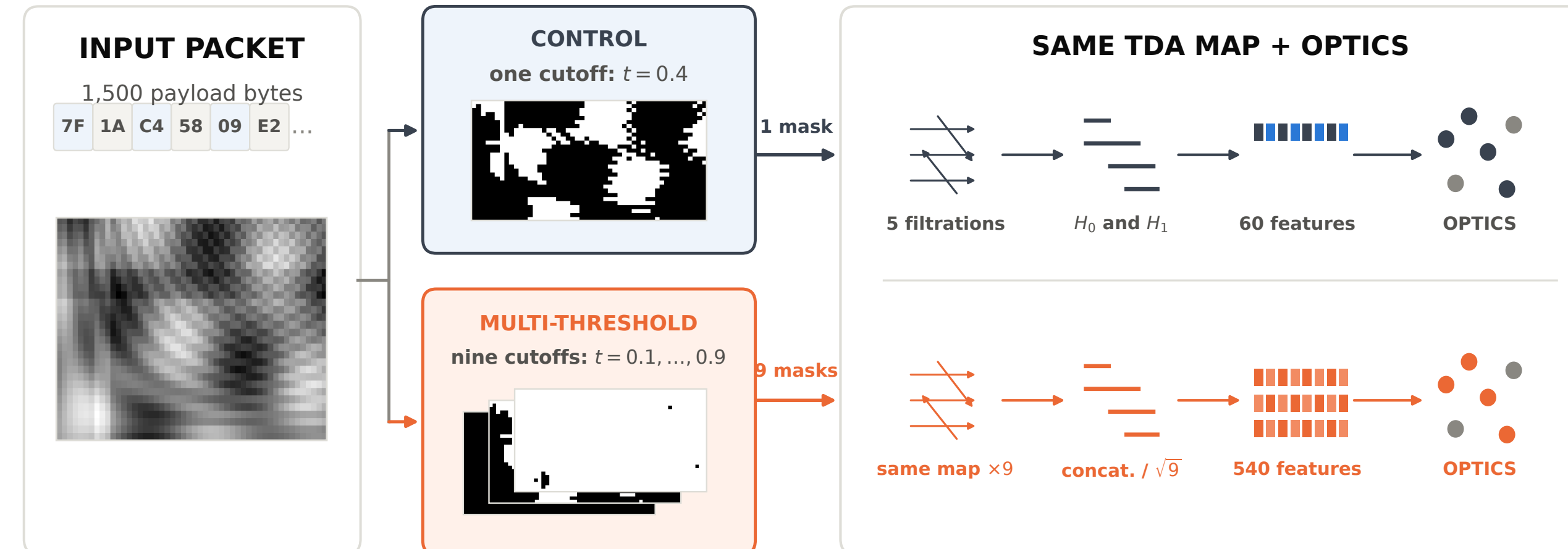
Reconstruction and Controlled Test

Data: UNSW-NB15 Payload-Byte [2].

1,500 bytes $\rightarrow$ 30×50 raster $\rightarrow$ $\text{Bin}_t \rightarrow$ 2 height + 3 radial filtrations $\rightarrow$ cubical persistence $(H_0, H_1) \rightarrow \mathbb{R}^{60} \rightarrow$ OPTICS

- Four probe families $\times$ five fixed seeds; each cell has 5,000 clean + 500 poison packets. Values = means $\pm$ population SD.
- 4 permutation families: Transpositions (Swap 60 pairs of random bytes), Block Reversal (Mirror 120-byte block), Block Swap (Swap two 60-byte blocks), Cyclic Shift (Rotate every row by 1 index).
- Every probe changes the raw vector and preserves its byte multiset.
- Ground-truth labels are used for scoring only.

TDA Pipeline



Results

92–96% Fewer Poisoned-Clean Vector Collisions using a multiple-threshold approach.

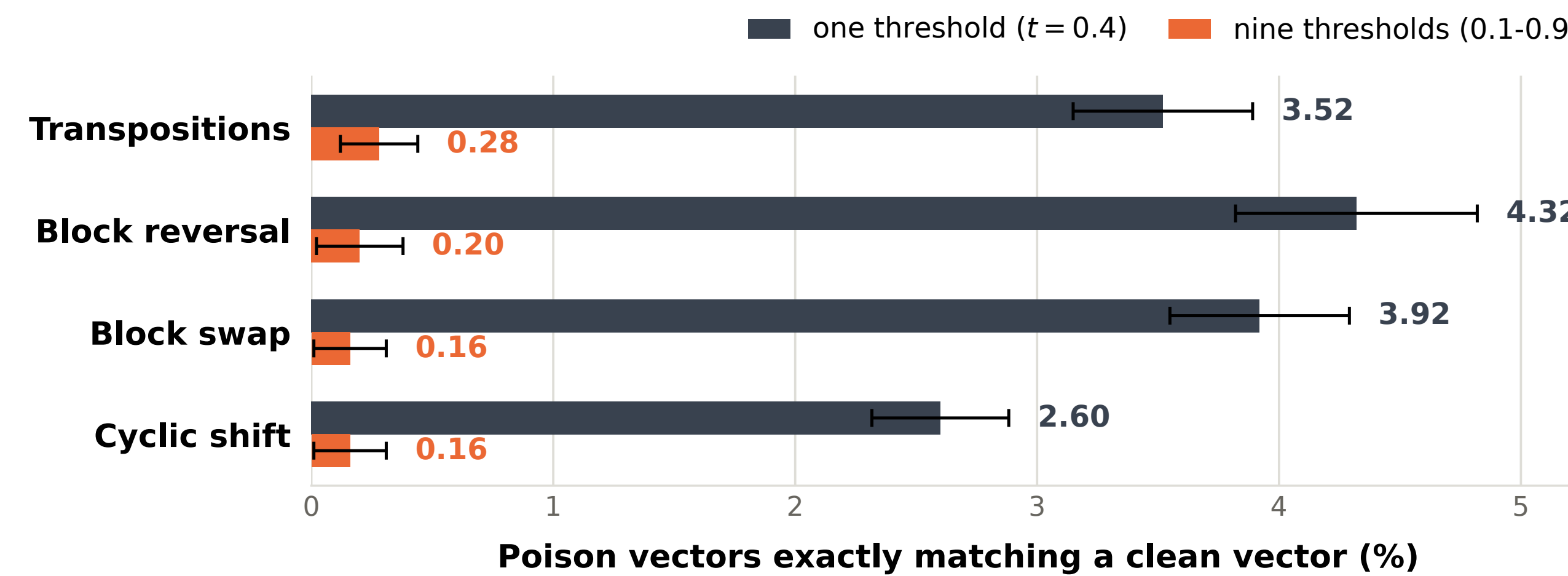


Figure 3. Means over five seeds; 5,000 clean + 500 poison packets per cell. Every probe changes the raw byte vector, so each collision counted here is one that binarization could in principle repair. Nine thresholds cut exact poison–clean matches by 92–96%.



Figure 4. Where collisions originate, from a separate audit of the single-threshold $\mathbb{R}^{60}$ pipeline. Half the collision mass is already present in the raw payload and is unreachable by any thresholding scheme; the 46.7% introduced at binarization is the share Figure 3 targets.

References and Contact

- [1] G. F. Monkam, M. J. De Lucia, and N. D. Bastian, *Computers & Security* 144, 103929, 2024.
- [2] N. Moustafa and J. Slay, “UNSW-NB15,” *MilCIS*, 2015.
- [3] M. Ankerst et al., “OPTICS,” *SIGMOD*, pp. 49–60, 1999.



Why One Cutoff Loses Information

One network packet (Payload Byte) from UNSW-NB15 is represented by $x \in \{0, \dots, 255\}^n$, $n = 1,500$.

The maximum byte value M is defined as $\max_i x_i = 255$; each byte is then normalized $z = x/M \in [0, 1]^n$.

Define the binarizer at threshold t by

$$\text{Bin}_t(z)_i = \begin{cases} 1, & z_i > t, \\ 0, & z_i \leq t. \end{cases}$$

Let $\sigma \in S_n$ permute byte positions. Since σ preserves the byte multiset it fixes M , so $\sigma z = (\sigma x)/M$ and

$$\text{Bin}_t(\sigma z) = \sigma \text{Bin}_t(z).$$

Original pipeline uses a single cutoff, $t = 0.4$:

$$A^- = \{i : z_i \leq 0.4\}, \quad A^+ = \{i : z_i > 0.4\}.$$

Any permutation that acts within these two classes leaves the binarization unchanged, so the stabilizer of z is contained in the stabilizer of its binarization:

$$G_z := \text{Stab}(z) \leq G_{0.4} := \text{Stab}(\text{Bin}_{0.4}(z)) = S(A^-) \times S(A^+).$$

$$\text{Blind spot: } G_{0.4} \setminus G_z.$$

Solution: 9 thresholds $T = \{0.1, \dots, 0.9\}$. Each position is sorted into ten bands $B_0, \dots, B_9$, so the stabilizer shrinks to

$$G_z \leq G_T := \text{Stab}(Q_T(z)) = \prod_{k=0}^9 S(B_k) \leq G_{0.4}.$$

Permutations within a single band still survive. Maps applied downstream remain noninjective.

Reducing representation invariance does not guarantee detection, but probability decreases substantially.

Conclusion and Scope

- Nine thresholds reduce exact poison–clean collisions by 92–96% among raw-changing poison packets, confirming that the single-cutoff blind spot is real and closable.
- Closing it does **not** improve detection. The 540-dimensional representation mainly fragments the OPTICS clustering, so poison packets are labeled noise rather than recovered as a cluster.
- Future work should improve attack separation relative to ordinary clean variation before retuning or replacing OPTICS.